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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/827,823	05/30/2022	Arnold MUELLER-RENTZ	818979	1802

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Leydig, Voit & Mayer, Ltd. (Frankfurt office)
Two Prudential Plaza, Suite 4900
180 North Stetson Avenue
Chicago, IL 60601-6731

EXAMINER	
BRIGGS, NATHANAEL R	

ART UNIT	PAPER NUMBER
2871	

NOTIFICATION DATE	DELIVERY MODE
03/18/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

chgpatent@leydig.com

Office Action Summary	Application No. 17/827,823	Applicant(s) MUELLER-RENTZ et al.	
	Examiner NATHANAEL R BRIGGS	Art Unit 2871	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 30 May 2022.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-21 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-21 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 30 May 2022 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 8/18/2022.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claim Objections

Claim 1 is objected to because of the following informalities: The word "positon" should be changed to "position". Appropriate correction is required.

Claim Rejections - 35 USC § 102

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(2) the claimed invention was described in a patent issued under section 151, or in an application for patent published or deemed published under section 122(b), in which the patent or application, as the case may be, names another inventor and was effectively filed before the effective filing date of the claimed invention.

Claim(s) 1, 3-13 and 16-21 are rejected under 35 U.S.C. 102(a)(2) as being anticipated by Ikeda et al. (US 2022/0244516).

Regarding claim 1, Ikeda discloses a microscope for imaging a sample by a transmitted light contrasting method (see figure 1, for instance), the microscope comprising: an objective lens holder (25) or changer configured to place an objective

lens (21) of a number of objective lenses onto an optical axis (Ax1) of the microscope, a lens system (10) for forming an intermediate image of an exit pupil of any one of the number of objective lenses (22) placed onto the optical axis, the intermediate image being formed at a respective conjugated plane conjugate to the exit pupil ([0040]), and a control device (65) configured for automatically positioning a modulation element (16) onto the optical axis (Ax1) at a position (P0) related to the respective conjugated plane ([0109]).

Regarding claim 3, Ikeda discloses the microscope according to claim 1, wherein the control device (65) is connected to the objective lens holder (25) or changer for receiving information on the respective objective lens (22) placed onto the optical axis (Ax1).

Regarding claim 4, Ikeda discloses the microscope according to claim 1, wherein the control device (65) is connected to a focus drive ([0173]) of the microscope, the focus drive being configured to move the respective objective lens (22) placed onto the optical axis (Ax1) in an axial direction (Δz) parallel to the optical axis, the control device (65) is configured to receive information of an axis position of the respective objective lens ([0173]).

Regarding claim 5, Ikeda discloses the microscope according to claim 1, wherein the control device (65) is configured to axially move the modulation element (16) only if an axial position displacement of the respective conjugated plane (P0) exceeds a pre-determined threshold.

Regarding claim 6, Ikeda discloses the microscope according to claim 1, wherein the control device (65) is configured for automatically moving the modulation element (16) in a lateral direction (via turret, [0109]) with respect to the optical axis.

Regarding claim 7, Ikeda discloses the microscope according to claim 6, wherein the control device (65) is configured to determine a lateral displacement of the modulation element (16) according to a quantitative image quality analysis ([0109]).

Regarding claim 8, Ikeda discloses the microscope according to claim 7, wherein the quantitative image quality analysis ([0109]) comprises at least one of a quantitative reproducibility of an image impression or a quantitative reproducibility of a contrast transfer function ([0062]).

Regarding claim 9, Ikeda discloses the microscope according to claim 1, wherein the control device (65) is configured to exchange a first modulation element (16) by a second modulation ([0109]: “Note that an element changing part capable of selecting and changing the illumination side modulation element 16 to be arranged at the position P0 of the illumination pupil from the plurality of illumination side modulation elements 16”) element by a lateral movement (via, turret, [0109]) of the first modulation element (16) out of the optical axis and by a lateral movement of the second modulation element onto the optical axis (Ax1).

Regarding claim 10, Ikeda discloses the microscope according to claim 9, wherein the control device (65) is further configured to adapt an illumination (5) of the microscope according to the second modulation element (16).

Regarding claim 11, Ikeda discloses the microscope according to claim 1, wherein the microscope comprises an illumination unit (5) for generating an illumination

beam path for illuminating the sample (SA) to be imaged, and wherein the respective objective lens (22) placed onto the optical axis (Ax1) generates an imaging beam path for imaging the sample.

Regarding claim 12, Ikeda discloses the microscope according to claim 11, wherein the modulation element (16) comprises a transparent section arranged such that upon a pre-determined lateral displacement of the modulation element out of its position on the optical axis ([0040]), the imaging beam path passes the transparent section of the modulation element (16).

Regarding claim 13, Ikeda discloses the microscope according to claim 12, wherein the control unit (65) is configured to laterally move the modulation element (16) by the pre-determined lateral displacement in order to inactivate the modulation element (via turret, [0109]).

Regarding claim 16, Ikeda discloses the microscope according to claim 11, wherein the illumination unit (5) comprises a condenser lens aperture changer (14) for generating a respective collimated illumination beam path of a respective aperture, the control device (65) being connected to the condenser lens aperture changer (14) in order to illuminate the sample (SA) with a pre-determined spatial illumination spectrum.

Regarding claim 17, Ikeda discloses the microscope according to claim 16, wherein the condenser lens aperture changer (14) comprises a condenser lens (15) and a condenser aperture diaphragm changer (14).

Regarding claim 18, Ikeda discloses the microscope according to claim 1, wherein the microscope further comprises a digital image detection module (60) for detecting an image of the sample (SA) to be imaged.

Regarding claim 19, Ikeda discloses the microscope according to claim 18, wherein the control device (65) is connected to the digital image detection module (60) in order to detect intensity images for at least one of: displaying, documentation, quantitative reconstruction of specimen phase modulation, or quantitative reconstruction of specimen amplitude modulation ([0052]).

Regarding claim 20, Ikeda discloses the microscope according to claim 1, wherein the control device (65) is configured to determine at least one of a phase transfer function or an amplitude transfer function of the optical system of the microscope in order to reconstruct a quantitative phase image of the sample ([0068]).

Regarding claim 21, Ikeda discloses the microscope according to claim 1, further comprising a modulation element positioning device comprising at least one slider ([0109]) configured to carry the modulation device (16), the at least one slider configured to move in at least one of a direction parallel to or a direction lateral with respect to the optical axis (Ax1).

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have

> been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

Claim(s) 2 is rejected under 35 U.S.C. 103 as being unpatentable over Ikeda in view of Kaneki et al. (US 2014/0118820).

Regarding claim 2, Ikeda discloses the microscope according to claim 1. However, Ikeda does not expressly disclose wherein the control device is configured for automatically moving the modulation element in an axial direction parallel to the optical axis.

Kaneki discloses a microscope (see figure 2, for instance), wherein the control device (6) is configured for automatically moving ([0076]) the modulation element (1811) in an axial direction (Z Direction) parallel to the optical axis (XA).

It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to use the automatic control of the modulation element in an axial direction as Kaneki in the device of Ikeda. The motivation for doing so would have been to adjust the compensation of the compensator within a range of the P0 position of Ikeda by causing the driving unit to drive the compensator to improve flexibility of the functionality of the device while also increasing efficiency of the device by minimizing optimization times, as taught by Kaneki ([0012]; [0108]).

Claim(s) 14 is rejected under 35 U.S.C. 103 as being unpatentable over Ikeda in view of Saito et al. (US 2009/0040601).

Regarding claim 14, Ikeda discloses the microscope according to claim 1. However, Ikeda does not expressly disclose wherein the modulation element is embedded in an anti-reflection coated plane-parallel plate.

Saito discloses a microscope (see figure 1, for instance), wherein the modulation element is embedded in an anti-reflection coated plane-parallel plate ([0070]).

It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to embed the modulation element in an anti-reflection coating layer as Saito in the device of Ikeda. The motivation for doing so would have been to enable noise to be reduced further and makes it possible to acquire a bright polarization image, as taught by Saito ([0070]).

Claim(s) 15 is rejected under 35 U.S.C. 103 as being unpatentable over Ikeda in view of Hoffman (US 4,200,353).

Regarding claim 15, Ikeda discloses the microscope according to claim 1. However, Ikeda does not expressly disclose wherein the modulation element is embedded in an anti-reflection coated plane-parallel plate.

Hoffman discloses a microscope (see figure 1, for instance), wherein the modulation element (8) is tilted relative to the optical axis (column 3, lines 34-39).

It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to provide tilt to the modulation element as Hoffman in the device of Ikeda. The motivation for doing so would have been to enable noise to provide increased control over the extent of modulation contrast, as taught by Hoffman (column 3, lines 34-39).

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to NATHANAEL R BRIGGS whose telephone number is (571)272-8992. The examiner can normally be reached Monday - Friday, 9:00 am - 5:00 pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Michael Caley can be reached on (571)-272-2286. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/NATHANAEL R BRIGGS/
Primary Examiner, Art Unit 2871
3/11/2025